M30/M20 CAN 2.0 Message Definition

English version merged from CAN message definition and CAN test command documents

Message Definition

1. Time - CAN ID (dec): 800

Test command: `LOG CCOM1 CANTIMEB ONTIME 1`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	WeekTime	16	1	week	GPS week	0	uint_16
16	GpsTime	32	0.001	s	GPS time of week	0	uint_32

2. IMU Raw Angular Rate - CAN ID (dec): 801

Test command: `LOG CCOM1 CANRATERAWB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AngRateRawX	20	0.01	deg/s	X-axis angular rate	0	int_20
20	AngRateRawY	20	0.01	deg/s	Y-axis angular rate	0	int_20
40	AngRateRawZ	20	0.01	deg/s	Z-axis angular rate	0	int_20

3. IMU Raw Acceleration - CAN ID (dec): 802

Test command: `LOG CCOM1 CANACCRAWB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AccelRawX	20	0.0001	g	X-axis acceleration	0	int_20
20	AccelRawY	20	0.0001	g	Y-axis acceleration	0	int_20
40	AccelRawZ	20	0.0001	g	Z-axis acceleration	0	int_20

4. INS Positioning Status - CAN ID (dec): 803

Test command: `LOG CCOM1 CANISNSTATEB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	system_state	8	1		0: initialization; 1: GNSS mode; 2: integrated navigation mode; 3: pure inertial mode	0	uint_8
8	GpsNumSatsUsed	8	1		Number of satellites used by the main antenna	0	uint_8
16	satellite_status	8	1		0: no positioning/no heading; 1: SPP positioning and heading; 2: pseudorange differential positioning and heading; 3: combined dead reckoning; 4: RTK fixed positioning and heading; 5: RTK float positioning and heading; 6: SPP positioning without heading; 7: pseudorange differential positioning without heading; 8: RTK fixed positioning without heading; 9: RTK float positioning without heading	0	uint_8
24	GpsNumSats2Used	8	1		Number of satellites used by the secondary antenna	0	uint_8
32	GpsAge	16	0.01	s	Differential age	0	uint_16
48	GpsNumSats	8	1		Number of satellites tracked by the main antenna	0	uint_8
56	GpsNumSats2	8	1		Number of satellites tracked by the secondary antenna	0	uint_8

5. Latitude and Longitude - CAN ID (dec): 804

Test command: `LOG CCOM1 CANLATLONB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	PosLat	32	1e-7	deg	Latitude	0	int_32
32	PosLon	32	1e-7	deg	Longitude	0	int_32

6. Altitude - CAN ID (dec): 805

Test command: `LOG CCOM1 CANALTIB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	PosAlt	32	0.001	m	Altitude	0	int_32

7. Position Sigma - CAN ID (dec): 806

Test command: `LOG CCOM1 CANPOSSGMB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	PosESigma	20	0.0001	m	East position sigma	0	uint_20
20	PosNSigma	20	0.0001	m	North position sigma	0	uint_20
40	PosUSigma	20	0.0001	m	Up position sigma	0	uint_20

8. Geodetic Velocity - CAN ID (dec): 807

Test command: `LOG CCOM1 CANVELLB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	VelE	16	0.01	m/s	East velocity	0	int_16
16	VelN	16	0.01	m/s	North velocity	0	int_16
32	VelU	16	0.01	m/s	Up velocity	0	int_16
48	Vel	16	0.01	m/s	Ground speed	0	int_16

9. Geodetic Velocity Sigma - CAN ID (dec): 808

Test command: `LOG CCOM1 CANVELLSGMB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	VelESigma	16	0.001	m/s	East velocity sigma	0	uint_16
16	VelNSigma	16	0.001	m/s	North velocity sigma	0	uint_16
32	VelUSigma	16	0.001	m/s	Up velocity sigma	0	uint_16
48	VelSigma	16	0.001	m/s	Ground speed sigma	0	uint_16

10. Vehicle-Frame Acceleration - CAN ID (dec): 809

Test command: `LOG CCOM1 CANACCVEHB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AccelX	20	0.0001	g	X-axis acceleration	0	int_20
20	AccelY	20	0.0001	g	Y-axis acceleration	0	int_20
40	AccelZ	20	0.0001	g	Z-axis acceleration	0	int_20

11. Heading, Pitch, and Roll - CAN ID (dec): 810

Test command: `LOG CCOM1 CANHPRB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AngleHeading	16	0.01	deg	Heading angle	0	uint_16
16	AnglePitch	16	0.01	deg	Pitch angle	0	int_16
32	AngleRoll	16	0.01	deg	Roll angle	0	int_16

12. Heading, Pitch, and Roll Sigma - CAN ID (dec): 811

Test command: `LOG CCOM1 CANHPRSGMB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AngleHeadingSigma	20	0.0001	deg	Heading angle sigma	0	uint_20
20	AnglePitchSigma	20	0.0001	deg	Pitch angle sigma	0	uint_20
40	AngleRollSigma	20	0.0001	deg	Roll angle sigma	0	uint_20

13. Vehicle-Frame Angular Rate - CAN ID (dec): 812

Test command: `LOG CCOM1 CANRATEVEHB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	AngRateX	20	0.01	deg/s	X-axis angular rate	0	int_20
20	AngRateY	20	0.01	deg/s	Y-axis angular rate	0	int_20
40	AngRateZ	20	0.01	deg/s	Z-axis angular rate	0	int_20

14. Longitude - CAN ID (dec): 813

Test command: `LOG CCOM1 CANLONB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	PosLon2	64	1e-8	deg	Longitude	0	int_64

15. Latitude - CAN ID (dec): 814

Test command: `LOG CCOM1 ANLATB ONTIME 0.01`

Offset (bit)	Signal	Length (bit)	Scale	Unit	Description	Offset	Format
0	PosLat2	64	1e-8	deg	Latitude	0	int_64